FL Benchmark Dashboard

ULB CREDIT CARD FRAUD · MLP MODEL · 5 ALGORITHMS × 3 PRIVACY MODES · 60 ROUNDS

FedAvg FedProx SCAFFOLD FedNova PersFL

PRIVACY MODE

All | No DP | DP | Sparse

BEST AUC	AVG BEST F1	FASTEST RUN	AVG JFI
0.9871	0.477	44.5s	0.9964
FedNova · DP	across 15 configs	FedNova · NoDP	Jain Fairness Index

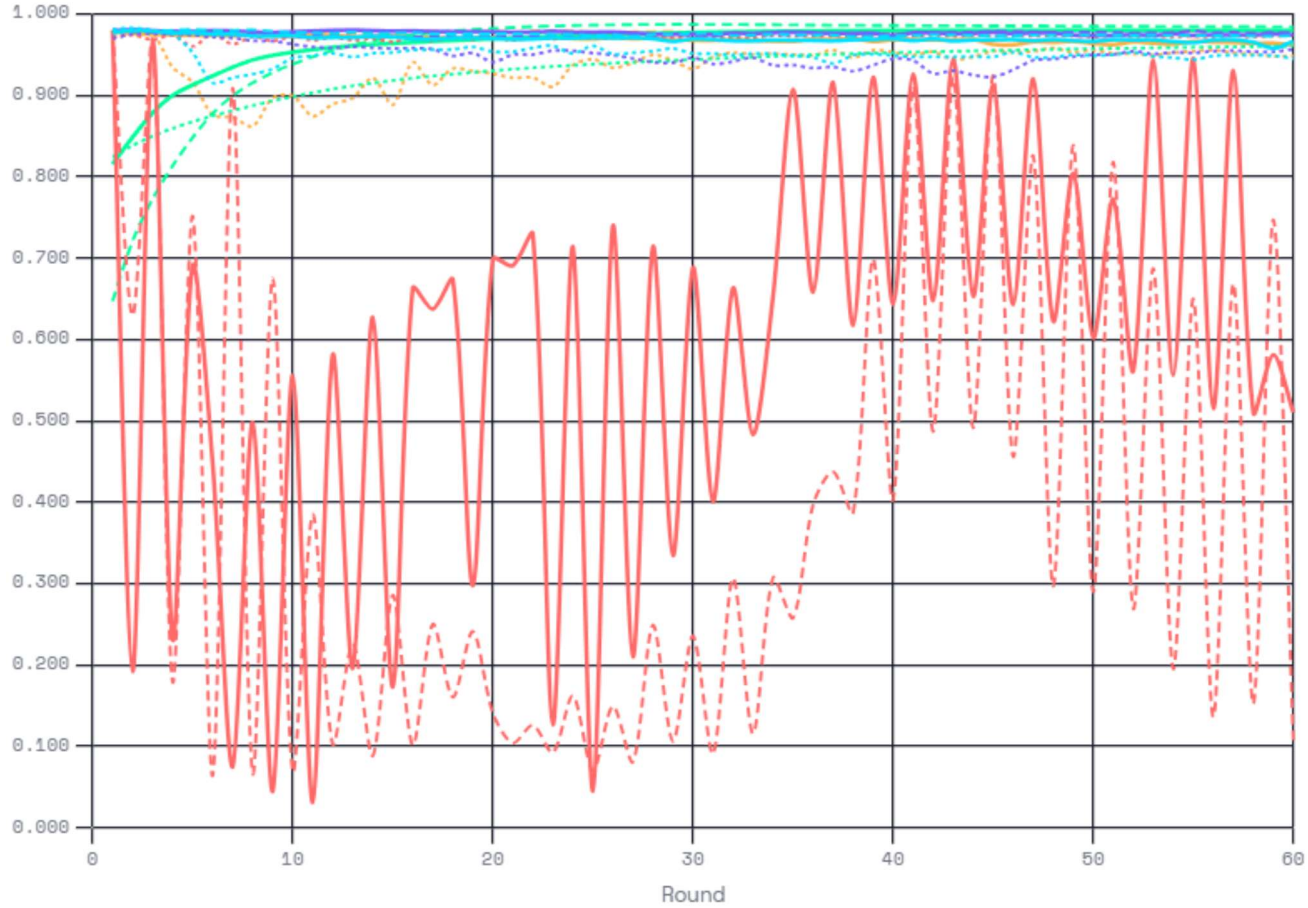
CONFIGS

15

filtered rows

AUC OVER TRAINING ROUNDS

AUC | F1 Score | Loss



FedAvg FedProx SCAFFOLD FedNova PersFL

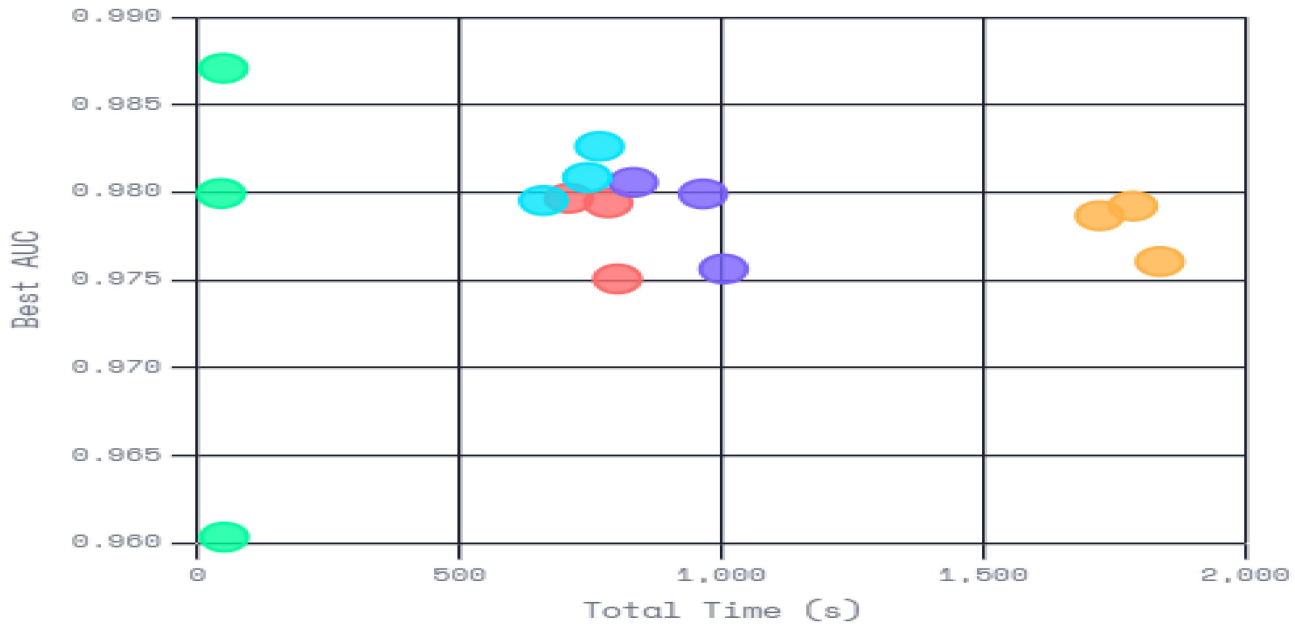
BEST AUC HEATMAP — ALGORITHM × PRIVACY MODE

	NO DP	DP	SPARSE
FedAvg	0.9795 F1: 0.700	0.9808 F1: 0.656	0.9826 F1: 0.728
FedProx	0.9806 F1: 0.361	0.9799 F1: 0.620	0.9756 F1: 0.244
SCAFFOLD	0.9797 F1: 0.720	0.9794 F1: 0.743	0.9751 F1: 0.038
FedNova	0.9799 F1: 0.185	0.9871 F1: 0.335	0.9604 F1: 0.009
PersFL	0.9787 F1: 0.585	0.9792 F1: 0.636	0.9761 F1: 0.594

PERFORMANCE RANKING TABLE

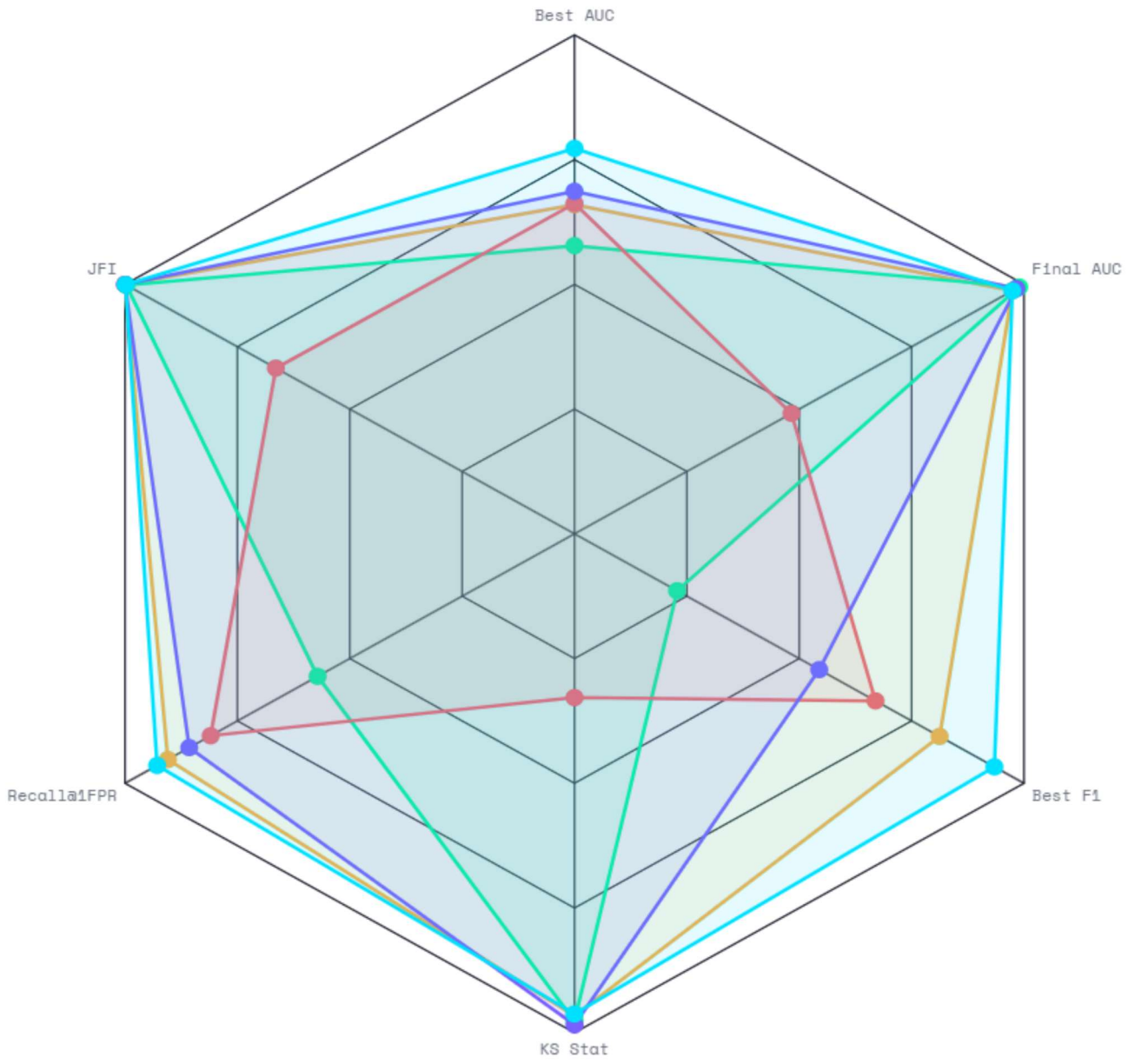
#	ALGORITHM	MODE	BEST AUC	FINAL AUC	BEST F1	KS STAT	TIME (S)	AUC BAR
1	FedNova	DP	0.9871	0.9841	0.335	0.903	49	
2	FedAvg	Sparsification	0.9826	0.9448	0.728	0.829	767	
3	FedAvg	DP	0.9808	0.9731	0.656	0.900	744	
4	FedProx	NoDP	0.9806	0.9741	0.361	0.898	831	
5	FedNova	NoDP	0.9799	0.9798	0.185	0.890	45	
6	FedProx	DP	0.9799	0.9768	0.620	0.904	965	
7	SCAFFOLD	NoDP	0.9797	0.5102	0.720	0.020	708	
8	FedAvg	NoDP	0.9795	0.9663	0.700	0.883	660	
9	SCAFFOLD	DP	0.9794	0.1053	0.743	0.013	783	
10	PersFL	DP	0.9792	0.9745	0.636	0.903	1785	
11	PersFL	NoDP	0.9787	0.9618	0.585	0.882	1721	
12	PersFL	Sparsification	0.9761	0.9483	0.594	0.838	1836	
13	FedProx	Sparsification	0.9756	0.9566	0.244	0.868	1004	
14	SCAFFOLD	Sparsification	0.9751	0.9732	0.038	0.883	801	
15	FedNova	Sparsification	0.9604	0.9604	0.009	0.858	51	

AUC VS TRAINING TIME



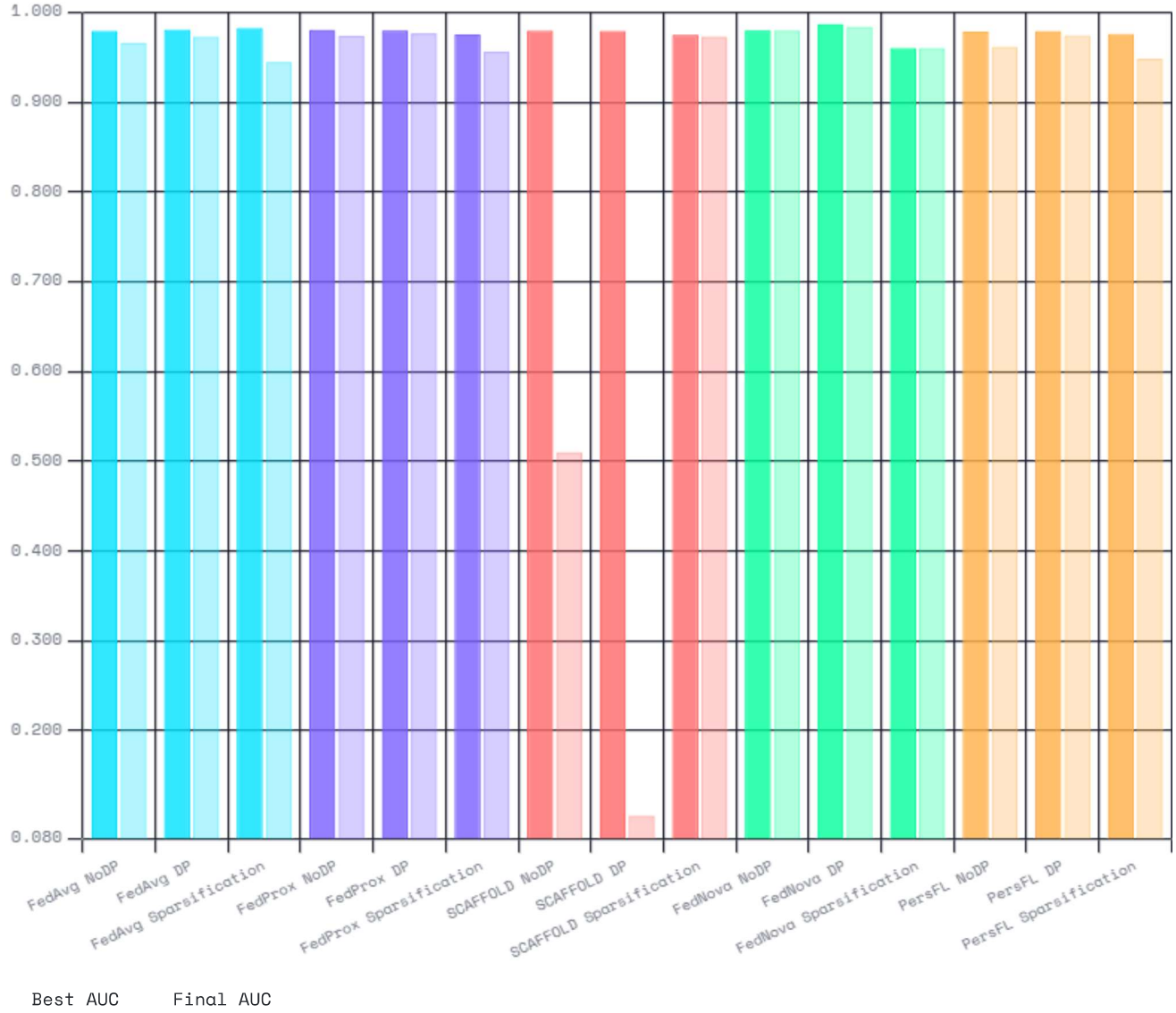
FedAvg FedProx SCAFFOLD FedNova PersFL

RADAR — ALGORITHM PERFORMANCE PROFILE



FedAvg FedProx SCAFFOLD FedNova PersFL

PRIVACY COST — BEST VS FINAL AUC


1.000
0.900
0.800
0.700
0.600
0.500
0.400
0.300
0.200
0.080
FedAvg NoDP
FedAvg DP
FedAvg Sparsification
FedProx NoDP
FedProx DP
FedProx Sparsification
SCAFFOLD NoDP
SCAFFOLD DP
SCAFFOLD Sparsification
FedNova NoDP
FedNova DP
FedNova Sparsification
PersFL NoDP
PersFL DP
PersFL Sparsification
Best AUC
Final AUC